FIITJEE Students From All Programs have bagged 34 in Top 100, 66 in Top 200 and 174 in Top 500 All India Ranks. FIITJEE Performance in JEE (Advanced), 2014:
2521 FIITJEE Students from Classroom / Integrated School Programs & 3579 FIITJEE Students from All Programs have qualified in JEE (Advanced), 2014.

FIITJEE

JEE(Main)-2015

ALL INDIA TEST SERIES

ANSWERS, HINTS & SOLUTIONS

CRT –IV
(Main)

S. No.	PHYSICS	CHEMISTRY	MATHEMATICS
1.	**A**	**B**	**A**
2.	**B**	**D**	**D**
3.	**D**	**A**	**A**
4.	**B**	**D**	**A**
5.	**C**	**B**	**C**
6.	**C**	**C**	**C**
7.	**B**	**C**	**C**
8.	**D**	**D**	**D**
9.	**D**	**B**	**B**
10.	**A**	**C**	**A**
11.	**D**	**D**	**B**
12.	**B**	**C**	**C**
13.	**A**	**B**	**A**
14.	**C**	**B**	**A**
15.	**C**	**A**	**A**
16.	**C**	**C**	**A**
17.	**B**	**C**	**C**
18.	**D**	**A**	**B**
19.	**A**	**B**	**B**
20.	**C**	**B**	**B**
21.	**B**	**A**	**B**
22.	**B**	**C**	**B**
23.	**D**	**C**	**B**
24.	**D**	**B**	**A**
25.	**C**	**D**	**C**
26.	**C**	**C**	**B**
27.	**A**	**D**	**C**
28.	**A**	**A**	**D**
29.	**D**	**C**	**C**
30.	**A**	**D**	**C**

Physics

PART – I

SECTION – A

1. Angular frequency of SHM is equal to angular velocity of particle $\omega = V/R$.

2. $V = n\lambda \quad \sqrt{\frac{T}{M}} = n(2\Delta x) \Rightarrow \sqrt{T} \propto \Delta x$

If tension is twice then Δx will increase to $\sqrt{2}\ \Delta x$.

3. $f_0 = \frac{1}{2l}\sqrt{\frac{T}{M}}$ [M = mass/length]

for elastic wire $T = Kx$

$x \rightarrow$ elongation from natural length

Hence frequency depends on K and original tension.

5. Temperature is maximum for which product of P and V is maximum as no. of moles is same for each containers

$PV = nRT$

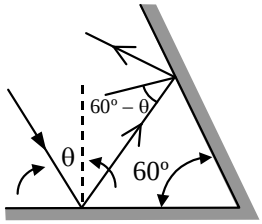
$T \propto PV$

Hence option (c) is true.

6. $S = [\pi - 2\theta] + [\pi - 2(60 - \theta)]$

$= 2\pi - 2\theta - 120 + 2\theta = \text{constant}$

Angle of incidence and final reflected ray does not depend on θ.



8. It only depends on temperature. In a mixture all gases has same temperature.

9. $\Delta U = nC_V\Delta T$ From graph

Maximum change in internal [$T_3 > T_2 > T_1$] energy is in process 4.

10. $n = \left(1 - \frac{300}{850}\right) = \frac{11}{17}$

Work done in each cycle is

$n = \frac{W}{Q_{in}} \quad Q_{in} = \frac{1200}{11} \times 17$

$Q_{rej} = Q_{in} - W = \frac{1200 \times 17}{11} - 1200$

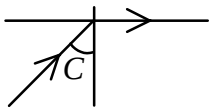
$= \frac{1200 \times 6}{11} = 654.54$

11. Quantities in option *a*, *b*, *c* has dimensions of $[M^1 L^{-1}T^2]$ but angular momentum per unit mass has dimension $[M^0 L^2 T^{-1}]$.

12. $\frac{\Delta R}{R}\times 100 = \frac{\Delta V}{V}\times 100 + \frac{\Delta I}{I}\times 100$

$= \frac{5}{100}\times 100 + \frac{0.2}{10}\times 100$

Total percentage error = 7%

13. 

$S = 2\pi - \left(\frac{\pi}{2} + C\right)$

$= \frac{\pi}{2} - C$

14. Series of resonant frequency must be integral multiple of fundamental frequency. This series is integral multiple of 75 H_2. Hence missing frequency is 75 H_2.

15. $V = \sqrt{\frac{T}{\mu}}$ $V \propto \sqrt{T}$

String with maximum tension will have maximum velocity.

V_1 = 3 m/sec, V_2 = 2m/s, V_3 = 4m/s

$V_4 = \frac{1}{2}$ m/s. Hence third string will have maximum tension.

16. $\frac{dQ}{dt} = KA\frac{\Delta\theta}{\Delta x}$

Rate of heat transfer, cross section area and thickness is same for all slab therefore

$K \propto \frac{1}{\Delta\theta}$

Slab which has minimum difference of temperature across interface will have maximum thermal conductivity.

Hence (c) option is true.

19. The number of mother nuclei decaying in a short time interval dt is = $N_m\lambda_m dt$. But death of a mother nucleus implies the birth of a daughter nucleus.

The number of daughter nuclei decaying in the same time interval is = $N_d\lambda_d dt$.

The number of daughter nuclei will remain constant when $N_m\lambda_m dt = N_d\lambda_d dt$

or $N_m\lambda_m = N_d\lambda_d$

21. Let fission rate be n per second; then

$n \times 200 \times 10^6 \times 1.6 \times 10^{-19} = 5 \times 1$

$\therefore \quad n = \frac{5}{3.2\times 10^{-11}} = 1.56\times 10^{11}$ / sec

22. $I = \frac{2}{80} = 25$ mA

24. The acceleration of particle at $t = 0$ is 0.

25. After the upper spring is cut, the block 1 and 2 will move with same acceleration.

$\therefore \quad a = 2g$

26. $\frac{dV}{dt} = (Q - av)$

$\frac{Adh}{dt} = (Q - av)$

$\int_{h_1}^{h_2} \frac{Adh}{Q - K\sqrt{h}} = \int_0^t dt$

$t = \frac{2A}{K^2}\left[K\left(\sqrt{h_1} - \sqrt{h_2}\right) - Q\ln\left(\frac{Q - K\sqrt{h_2}}{Q - K\sqrt{h_1}}\right)\right]$

27. $\frac{1}{2}KA^2 + \frac{1}{2}2m\frac{V_0^2}{4} = \frac{1}{2}KA'^2$ (by energy and momentum considerations)

$KA^2 + \frac{1}{2}mV_0^2 = KA'^2 \Rightarrow A' = \sqrt{2}A$

$A' = \sqrt{2}A \Rightarrow \varphi = \frac{5\pi}{4}$

$x = A'\sin(\omega' t + \varphi)$

$\omega' = \sqrt{\frac{K}{2m}}$

$x = -\sqrt{2}A\sin\left(\sqrt{\frac{K}{2m}}t + \frac{\pi}{4}\right)$

28. $\lambda_1 = 1085\text{Å}$ & $\lambda_2 = 304\text{Å}$

$E_1 = \frac{hc}{\lambda_1} = 11.5\text{eV}$ $E_2 = \frac{hc}{\lambda_2} = 40.8\text{eV}$

KE of electron after collision
= 100 – 52.3 = 47.7 eV

29. Net force on the system is zero hence momentum is conserved so
$mv_0 = 2mv$
Both wires will have same velocity when steady state will be achieved.

$\Delta K = \frac{1}{4}mv_0^2$

30. $T = 2\pi\sqrt{\frac{I}{mgl}} \Rightarrow f = \frac{1}{2\pi}\sqrt{\frac{mgl}{I}}$

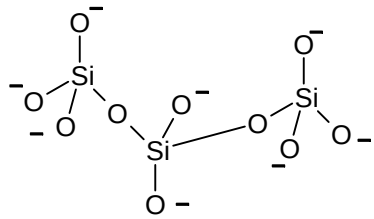
I is moment of inertia about axis passing from suspension point.

$f \propto \frac{1}{\sqrt{I}}$, $\in I$ about axis side to side in plane of loop is more hence *f* is greater in first case.

Chemistry

PART – II

SECTION – A

1. 

Structure of silicate anion $\left(Si_3O_{10}^{8-}\right)$ in mineral kinoite.

2. NO_2 is a brown coloured gas:

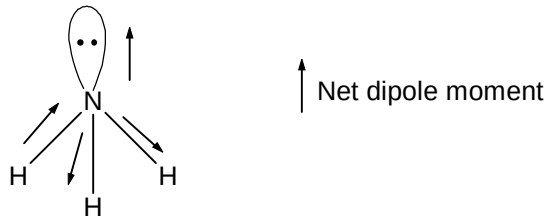
$$Ca(NO_3)_2 \xrightarrow{\Delta} CuO \downarrow +2NO_2 \uparrow +\frac{1}{2}O_2 \uparrow$$

$$Hg(NO_3)_2 \xrightarrow{\Delta} Hg \downarrow +2NO_2 \uparrow +O_2 \uparrow$$

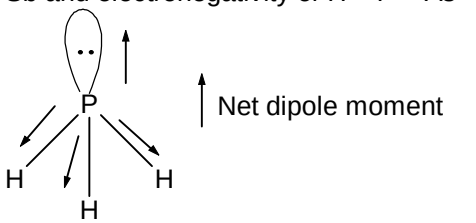
$$NaNO_3 \xrightarrow[\text{below } 500^oC]{\Delta} NaNO_2 \downarrow +\frac{1}{2}O_2 \uparrow$$

$$AgNO_3 \xrightarrow{\Delta} Ag \downarrow +NO_2 \uparrow +\frac{1}{2}O_2 \uparrow$$

3. Bond order of O_2^+ is 2.5
Bond order of NO is 2.5
Bond order of N_2^- is 2.5
Bond order can be calculated using molecular orbital diagrams.

4. 

Bond angle around central atom decrease down the group. Also electronegativity of N > P > As > Sb and electronegativity of H > P > As > Sb. Electronegative of P is very close that of H.



$Ca_3P_2 + 6H_2O \longrightarrow 3Ca(OH)_2 + 2PH_3$

$2F_2 + 2H_2O \longrightarrow 4HF + O_2$

$3F_2 + 3H_2O \longrightarrow 6HF + O_3$

5. XeF_6 may act as fluoride donor when it reacts with non metal fluorides.

$XeF_6 + SbF_3 \rightleftharpoons SbF_6^- + XeF_5^+$

$XeF_6 + \text{excess } H_2O \longrightarrow XeO_3 + 6HF$

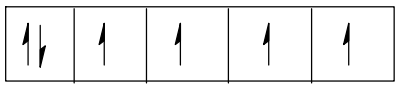
$XeF_6 + H_2O \longrightarrow XeOF_4 + 2HF$

6. $\underset{\downarrow\ x}{\left[Co\right.}\ \underset{\downarrow\ 0}{(NH_3)_5}\ \underset{\downarrow\ -2}{CO_3}\left.\right]\underset{\downarrow\ -1}{ClO_3}$

$x + 0 \times 5 + (-2) \times 1 = +1$

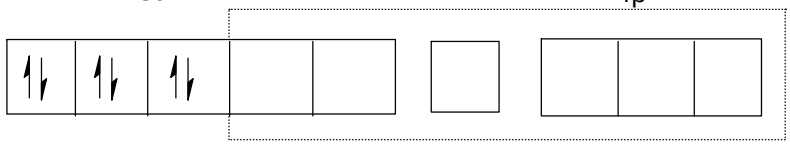
$x = +3$

Co^{3+}



3d 4s 4p

in Co^{3+} complex



d^2sp^3

Coordination number = 6

No. of unpaired e^- (n) = 0

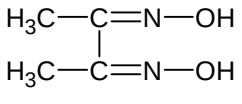
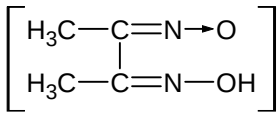
$\therefore\ \mu = \sqrt{n(n+2)} = 0$

Number of 'd' electron = 6

7. $Hg + X_2 \longrightarrow HgX_2$ (where X = F, Cl, Br or I)

Mac-Aurthor's process is used in extraction of gold and silver

$4M + 8NaCN + 2H_2O + O_2 \longrightarrow 4Na\left[M(CN)_2\right] + 4NaOH$ Where, M = Ag, Au.

8. $Ni^{2+} + NH_3 +$  $\longrightarrow$ Ni 

Red precipitate

$Hg^{2+} + Co(SCN)_2(aq) \longrightarrow \underset{\text{Blue precipitate}}{Hg(SCN)_2} + CO^{2+}$

$3Mn(NO_3)_2(aq) + 5NaBiO_3(aq) + 9HNO_3(aq) \longrightarrow \underset{(\text{Purple})}{3HMnO_4(aq)} + 3H_2O(\ell) + 5Bi(NO_3)_3$

9. Pd $\Rightarrow 4d^{10}5s^0 \Rightarrow 10\ \alpha$ electron

S $\Rightarrow 1s^2 2s^2 2p^6 3s^2 3p^4 \Rightarrow 10$ p electrons

10. Longest wavelength of Li^{2+}, Balmer series $\Rightarrow n_1 = 2, n_2 = 3$

$\frac{1}{x} = R_H \times 3^2\left[\frac{1}{2^2} - \frac{1}{3^2}\right]\ \left[\because \frac{1}{\lambda} = R_H Z^2\left[\frac{1}{n_1^2} - \frac{1}{n_2^2}\right]\right]$

$\frac{1}{x} = R_H \times 9\left[\frac{9-4}{36}\right]$ ....(1)

Shortest wavelength of He^+ Paschen series $\Rightarrow$

$n_1 = 3$, $n_2 = \infty$

$\frac{1}{\lambda} = R_H \times 2^2\left[\frac{1}{3^2} - \frac{1}{\infty^2}\right]$ .....(2)

From equation (1) and (2)

$\frac{\lambda}{x} = \frac{R_H \times 9 \times \frac{5}{36}}{R_H \times 4 \times \frac{1}{9}} = \frac{5 \times 9}{4 \times 4} = \frac{45}{16}$

$\lambda = \frac{45x}{16}$

11.

$V_{rms} = \sqrt{\frac{3RT}{M}}$ $V_{avg} = \sqrt{\frac{8RT}{\pi M}}$ $V_{m.p} = \sqrt{\frac{2RT}{M}}$

For N_2 at 27°C For H_2 at 27°C For CO at 27°C

$V_{rms} = \sqrt{\frac{3RT}{28}}$ $V_{avg} = \sqrt{\frac{8RT}{2\pi}}$ $V_{mp} = \sqrt{\frac{2RT}{28}}$

$V_{rms} : V_{avg} : V_{mp} = \sqrt{\frac{3}{28}} : \sqrt{\frac{4}{\pi}} : \sqrt{\frac{1}{14}}$

= 0.33 : 1.128 : 0.267

12. Mass of each atom = $\frac{\text{Atomic wt.}}{N_A} = \frac{109.75}{6.022 \times 10^{23}} = 1.82 \times 10^{-22}$

In FCC number of atom/unit cell = 4

Mass of unit cell = $4 \times 1.82 \times 10^{-22} = 7.29 \times 10^{-22}$ g

$9\text{g/cm}^3 = \text{density} = \frac{\text{mass}}{\text{volume}} = \frac{7.29 \times 10^{-22}\text{g}}{a^3}$

$a^3 = \frac{7.29 \times 10^{-22}\text{g}}{9\text{g/cm}^3} = 81 \times 10^{-24}\text{cm}^3$

a = 900 pm

13. $A(OH)_2 \rightleftharpoons A^{2+} + 2OH^-$

$\frac{0.0416\text{g/L}}{104\text{g/mol}}$

$\left[A^{2+}\right] = 4 \times 10^{-4}\text{g/mol}$

$\left[OH^-\right] = 10^{-pOH} = 10^{-3}$

$K_{sp} = \left[A^{2+}\right]\left[OH^-\right]^2$

$= \left(4 \times 10^{-4}\right)\left(10^{-3}\right)^2 = 4 \times 10^{-10}$

$K_{sp} = 4s^3$ for $A(OH)_2$

$$S = \sqrt[3]{\frac{Ksp}{4}} = \sqrt[3]{\frac{4\times10^{-10}}{4}} = 2.16\times10^{-3}$$

14.

	$3A \rightleftharpoons$	$2B$	+	$2C$	+	D
Initial	P^o	0		0		0
Equilibrium	$P^o - 2x$	$2x$		$2x$		x

Equilibrium pressure = $P^o - 2x + 2x + 2x + x$

$50 = P^o + 3x$ ……..(1)

From question $5 = P^o - 2x$ ……..(2)

From equation (1) and (2)

$55 = 5x$

$x = 11$

$$K_p = \frac{P_B^2.P_C^2.P_D}{P_A^3} = \frac{(22)^2(22)^2 11}{5^3}$$

= 20614.5

$$K_p = K_c(RT)\Delta n$$

$$K_c = \frac{K_p}{(RT)^{\Delta n}} = \frac{20614.5}{(0.082\times400)^2}$$

= 19.16

15. Adiabatic reversible expansion

$$T_1^{\gamma}P_1^{1-\gamma} = T_2^{\gamma}P_2^{1-\gamma}$$

$$T_2^{\gamma} = T_1^{\gamma}\left(\frac{P_1}{P_2}\right)^{1-\gamma}$$

For mono atomic gas $\gamma = 1.66$

$T_2 = 172.88$

$T_2 = -100.27°C$

16. $$C_6H_{10}(\ell) + 8\frac{1}{2}O_2(g) \longrightarrow 6CO_2(g) + 5H_2O(\ell)$$

$\Delta H = \Delta E + \Delta ng\ RT$

$\Delta H = \Delta E + (6 - 8.5)\ RT$

$\Delta H = \Delta E - 2.5\ RT$

17. $$t = \frac{2.303}{k}\log\frac{a_o}{a_t}$$

$$k = \frac{0.693}{t_{1/2}} = \frac{0.693}{20}$$

$$t = \frac{2.303}{\frac{0.693}{20}}\times\log\frac{10}{10^{-7}}$$

= 34.75 min

18. Adsorption of gas on solid is always exothermic.

19. $m = \frac{\Delta T_b}{K_b} = \frac{0.104K}{0.52K\ Kg\ Mol^{-}} = 0.2\ mol\ Kg^{-1}$

$\pi = MRT \Rightarrow M = \frac{\pi}{RT}$

$= \frac{7.38\ atm}{\left(0.082\ L\ atm\ mol^{-1}\ K^{-1}\right)\left(300\ K\right)}$

$= 0.3\ mol\ L^{-1}$

Assuming volume of water does not change significantly on addition of glucose

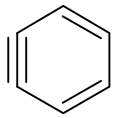
In 1 L solution $\Rightarrow$ 0.3 mol glucose and 1 kg water

$density = \frac{mass\ of\ solution}{volume\ of\ solution}$

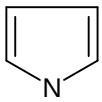
$= \frac{0.3\ mol \times \left(180g\ mol^{-1}\right) + 1000g}{1000\ mL}$

$= 1.054\ g\ mL$

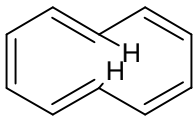
20.



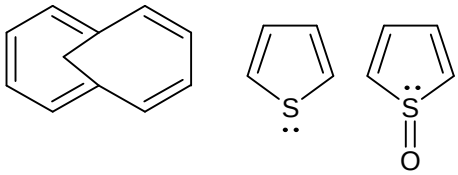
$8\pi e^{-}$. satisfying $(4n + 2)\pi e^{-}$ rule $\Rightarrow$ only $6\pi e^{-}$ are in one plane and in cyclic conjugate. Therefore, aromatic.



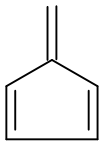
$6\pi e^{-}$ satisfying $(4n+2)\pi e^{-}$ rule $\Rightarrow$ aromatic



Due to repulsion between internal hydrogen atom $\Rightarrow$ non aromatic

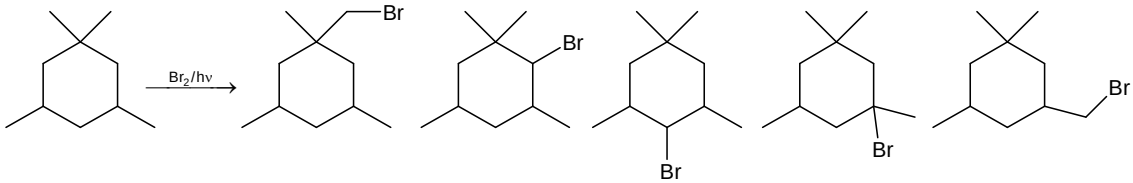


$10\pi e^{-}$ satisfies $(4n + 2)\pi e^{-}$ rule $\Rightarrow$ aromatic



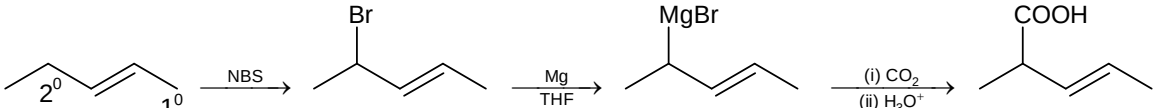
Non aromatic because cyclic delocalisation of πe^{-} is abscent.

21.

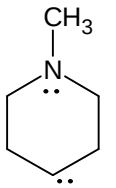


Five products

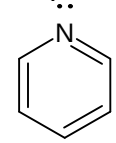
22.



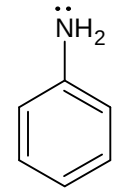
Allylic bromination

23. 

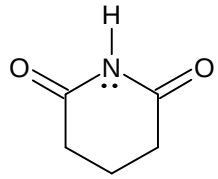
Aliphatic tertiary amine, nitrogen electron pair is not in conjugation.



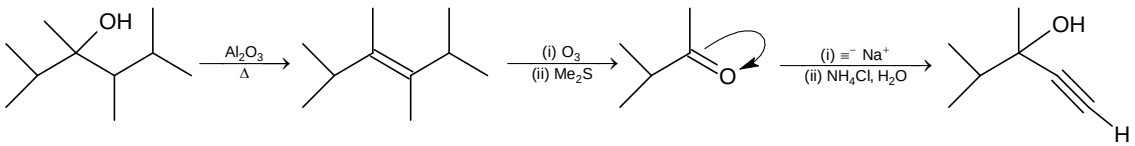
Nitrogen is sp^2 hybridised. Nitrogen electron pair is not in conjugation.

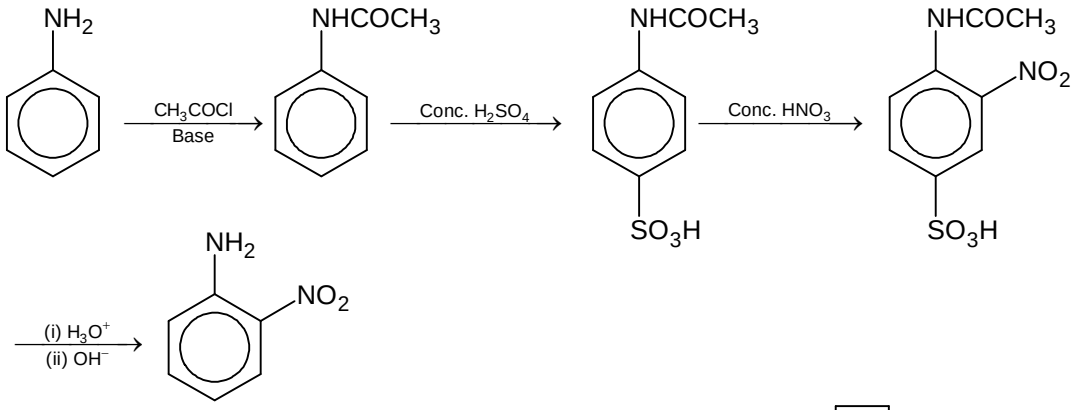


Nitrogen electron lone pair is in conjugation with phenyl ring.

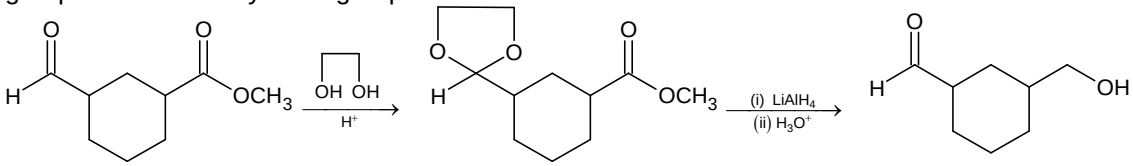


Nitrogen e^- lone pair is in conjugate with two C=O groups leading to two equivalent resonative structures.

24. 

25. 

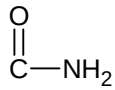
26. $LiAlH_4$ can reduce both aldehyde as well as ester group into. But use  protects aldehyde group and hence only ester group is reduced.

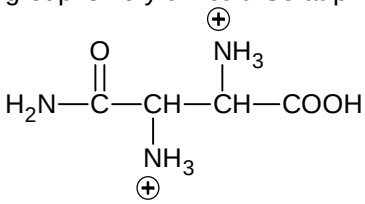


27. Hunsdieker reaction

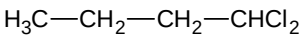
$$R-COOAg+Br_2 \longrightarrow R-Br+CO_2\uparrow+AgBr\downarrow$$

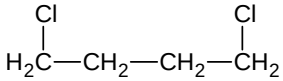
Free radical mechanism is followed.

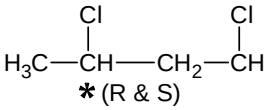
28. At pH = 3, condition is highly acidic so all NH_2 group will get protonated. Protonation of  group is very difficult. So at pH = 3 structure will be

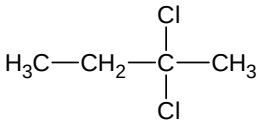


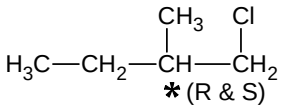
29.

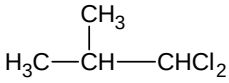


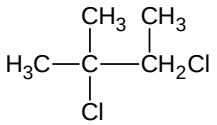


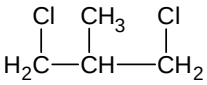












* Indicate chiral carbon

30. Balanced chemical equation is

$$2KMnO_4(aq)+5H_2C_2O_4(aq)+2HCl(aq) \rightleftharpoons 2MnO(s)+10CO_2(g)+6H_2O(\ell)+2KCl(aq)$$

K_p depends only on partial pressure of gaseous species

$$\therefore\ K_p=\left(P_{CO_2}\right)^{10}$$

Mathematics

PART – III

SECTION – A

1. $\lim\limits_{x\to 0}\dfrac{\sin(x-\sin x)}{\sin(x-\tan x)}=\lim\limits_{x\to 0}\dfrac{x-\sin x}{x-\tan x}=-\dfrac{1}{2}$

2. $\because f\left(\sqrt{3-x}\right)$ exists for $\sqrt{3-x}\geq 0$

   i.e. Domain of 'f' is $[0,\ \infty)$

   But, for $x>0\Rightarrow 0\leq\sqrt{3-x}<\sqrt{3}$

3. Suppose $|\alpha|<1$, we have

   $|-3|=|\alpha^5+\alpha^3+\alpha|\leq|\alpha|^5+|\alpha|^3+|\alpha|<1+1+1$

   $\Rightarrow 3<3$ A contradietion

   Thus $|\alpha|\geq 1\Rightarrow\alpha$ lies either on or outside the unit circle

4. Here $a>0, b>0, c>0, d>0$

   $\Rightarrow a^2+b^2+c^2\geq ab+bc+ca$

   $\Rightarrow 3(a^2+b^2+c^2)\geq(a+b+c)^2>d^2$

   $\Rightarrow\dfrac{a^2+b^2+c^2}{d^2}>\dfrac{1}{3}$

5. Clearly from Descarte's rule of sign equation has 4 positive real roots lets say α, β, γ, and δ

   $\Rightarrow\sum\alpha=-a_1;\ \sum\alpha\beta=-a_2;\ \sum\alpha\beta\gamma=-a_3$ and $\alpha\beta\gamma\delta=5$

   using AM $\geq$ GM $\Rightarrow\dfrac{\alpha+\beta+\gamma+\delta}{4}.\dfrac{\alpha\beta\gamma+\ldots}{4}\geq\sqrt[4]{\alpha\beta\gamma\delta}.\sqrt[4]{\alpha^3\beta^3\gamma^3\delta^3}=\alpha\beta\gamma\delta=5$

   $\Rightarrow-\dfrac{a_1}{4}.\dfrac{-a_3}{4}\geq 5\ \Rightarrow a_1a_3\geq 80$

8. Curve is $(y-2)^2=2x^3-4$ ..... (1)

   $\dfrac{dy}{dx}=\dfrac{3x^2}{y-2}$

   $\therefore$ Equation of tangent at (x_1, y_1)

   $y-y_1=\dfrac{3x_1^2}{y_1-2}(x-x_1)$

   As it passes through (1, 2)

   $\Rightarrow\ 2-y_1=\dfrac{3x_1^2}{y_1-2}(1-x_1)$

   or $(y_1-2)^2=3x_1^2(x_1-1)$

   or $2x_1^3-4=3x_1^3-3x_1^2$ [As (x_1, y_1) lies on the curve]

or $x_1^3 - 3x_1^2 + 4 = 0$
or $(x_1 + 1)(x_1 - 2)^2 = 0$
$\therefore x_1 = -1$ or 2
But $x_1 \neq = -1$ (it is clear from equation (1))
$\therefore x_1 = 2$
$\therefore y_1 = 2 \pm 2\sqrt{3}$
Points of contact are $(2, 2 + 2\sqrt{3})$ and $(2, 2 - 2\sqrt{3})$
$\therefore$ Sum of x–coordinates of points of contact is 4
$\therefore$ Sum of y–coordinates of points of contact is 4

9. If $\left|\int_a^b f(x)dx\right| = \int_a^b |f(x)|dx$

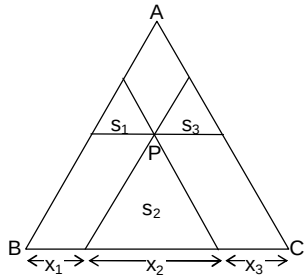
$\Rightarrow$ f(x) is either positive in (a, b)
or f(x) is negative in (a, b) and also b > a
Also g′(x) = f(x)
When f(x) > 0 $\Rightarrow$ g ′(x) > 0 $\Rightarrow$ g(x) is increasing function
When f(x) < 0 $\Rightarrow$ g ′(x) < 0 $\Rightarrow$ g(x) is decreasing function
Also g(0) = 0

$$\int_a^b f(x)g(x)dx = \int_a^b g(x)g'(x)dx$$

$$= \frac{(g(b))^2 - (g(a))^2}{2}$$ which will be positive for 0 < a < b

10. $\frac{s_1}{\Delta} = \frac{x_1^2}{a^2}$ or $\sqrt{\frac{s_1}{\Delta}} = \frac{x_1}{a}$

Similarly $\sqrt{\frac{s_2}{\Delta}} = \frac{x_2}{a}$ and $\sqrt{\frac{s_3}{\Delta}} = \frac{x_3}{a}$



11. Let $f^{-1}(x) = t \Rightarrow f(t) = x$

$$\int_{f(a)}^{f(b)} 2x\left(b - f^{-1}(x)\right)dx = \int_a^b 2f(t).(b-t).f'(t)dt = b\,[(f(t))^2]_a^b - \int_a^b t\,.2f(t)\,.f'(t)\,dt$$

$$= \int_a^b \left[\left(f(x)\right)^2 - \left(f(a)\right)^2\right]dx$$

12. $(1+z_0)\left(1+z_0^{2^1}\right)\left(1+z_0^{2^2}\right).\left(1+z_0^{2^n}\right) = \frac{1-z_0^{2^{n+1}}}{1-z_0} = \frac{1-\left(z_0^2\right)^{2^n}}{1-z_0} = (1+i)\left(1-\frac{1}{2^{2^n}}\right)$, if n > 1;

If n = 1, $(1+z_0)\left(1+z_0^2\right) = \frac{5}{4}(1+i)$

17. If one number is drawn from the set {1, 2, 3…….n} the probability α that it is divisible by 2 is given by $\alpha = \frac{\left[\frac{n}{2}\right]}{n}$

$\Rightarrow$ P (the number is not divisible by 2) = $1 - \alpha$
Now, $x^2 - y^2 = (x - y)(x + y)$
$\therefore P_2 = \alpha.\alpha + (1-\alpha)(1-\alpha) = 1 - 2\alpha + 2\alpha^2$

18. To exhaust all single digit numbers he must have written $\sum_{i=1}^{9} i^2 = 285$ digits, to exhaust 10 he must write 2×10^2 more digits i.e. 485 digits. So 500th digit will occur when he is writing 11

19. Taking $2^{\sin x} = t \Rightarrow \frac{1}{2} \le t \le 2$

$$a \le \frac{1-t^2}{t} \Rightarrow a \le \min\left(\frac{1-t^2}{t}\right) \Rightarrow a \le -\frac{3}{2}$$

20. If p_1 and p_2 are length of perpendiculars from foci to any tangent then $p_1p_2 = b^2 = 16$ and $p_1 + p_2 = 8 \Rightarrow p_1 = p_2 = 4$
Hence only two tangents are possible

21. Option (B) is correct because in this case $a > 0, b < 0$ (Let $b = -c$)

$$\Rightarrow -cx^2 + ay^2 = -ac \Rightarrow \frac{x^2}{a} - \frac{y^2}{c} = 1$$

22. $(\cos y - \sin y)\frac{dy}{dx} + (\cos y + \sin y)e^{-x} = e^{e^{-x}}$

$\cos y + \sin y = u$

$$\Rightarrow \frac{du}{dx} + e^{-x}u = e^{e^{-x}} \Rightarrow ue^{-e^{-x}} = x$$

$\Rightarrow e^{-e^{-t}} = t$ (Putting (t, 0))

$\Rightarrow t.e^{e^{-t}} = 1$

23. Fro more than one triplet satisfying all three equations we have

$$\begin{vmatrix} \frac{1}{a} & \frac{1}{b} & \frac{1}{c} \\ \frac{1}{b} & \frac{1}{c} & \frac{1}{a} \\ \frac{1}{c} & \frac{1}{a} & \frac{1}{b} \end{vmatrix} = 0$$

$\Rightarrow a = b = c$
$\Rightarrow (x + y + z) = a$ satisfying the triplets
We get $a = 10,\ p = 7 \Rightarrow \alpha^2 - 2\alpha - 8 = 0$
$\alpha = -2$ and 4

24. If $f(x)$ is continuous then $\lim_{k\to\infty} f(t_k) = f\left(\lim_{k\to\infty}(t_k)\right) = f\left(\frac{1}{4}\right) = 1$ (Where $\lim_{k\to\infty} t_k = \frac{1}{4}$)

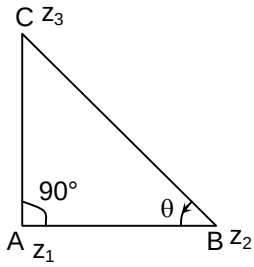
Also as graph is smooth $f'\left(\frac{1}{4}\right) = 0$

25. $f(x)=\dfrac{x-1}{x+3}\forall x\neq 2 \Rightarrow$ f(x) can not take the value 1 and f(2).

26. $\dfrac{W_2}{|W_2|}=\dfrac{W_1}{|W_1|}e^{-i\theta}$

$\Rightarrow \dfrac{W_2}{W_1}=\dfrac{|W_2|}{|W_1|}e^{-i\theta}=1-i\tan\theta$

$W_1\overline{W}_2=\dfrac{W_1}{W_2}|W_2|^2=|W_1|\ |W_2|e^{i\theta}=\left(|W_1|^2+i|W_1|\ |W_2|\sin\theta\right)$



27. $(13x-1)^2=36$

$13x-1=\pm 6$

$x=\dfrac{7}{13}$

$x=-\dfrac{5}{13}$

$\Rightarrow \cos\alpha=-\dfrac{5}{13}$

$\Rightarrow \sin\alpha=\pm\dfrac{12}{13}$

$\Rightarrow \sin 2\alpha=\pm\dfrac{120}{169}$

28. $(1+x+x^2)^n=a_0+a_1x+a_2x^2+\ldots+a_{2n}x^{2n}$ ..... (1)

$\Rightarrow a_0+a_2+a_4\ldots=\dfrac{3^n+1}{2}$

$a_1+a_3+a_5\ldots=\dfrac{3^n-1}{2}$

Put $x=i$ in equation (1) where $i=\sqrt{-1}$

If $n=4m+1$

$\Rightarrow a_1-a_3+a_5\ldots=1$

$a_0-a_2+a_4\ldots=0$

So, $a_1+a_5+a_9\ldots=\dfrac{3^n+1}{4}=S_2$

$a_3+a_7+a_{11}\ldots=\dfrac{3^n-3}{4}=S_4$

Similarly if $n=4m+3$

Then, $S_2=\dfrac{3^n-3}{4}$ and $S_4=\dfrac{3^n+1}{4}$

29. If n is even

$f(n)=2\left(1+2+\ldots\dfrac{n-2}{2}\right)+\dfrac{n}{2}=\dfrac{n^2}{4}$

If n is odd then $f(n)=f(n-1)+\dfrac{n-1}{2}=\dfrac{(n-1)^2}{4}+\dfrac{n-1}{2}=\dfrac{n^2-1}{4}$

$f(n+m)-f(n-m)=\dfrac{(n+m)^2-(n-m)^2}{4}$ if both even

$=\dfrac{(n+m)^2-1-(n-m)^2+1}{4}$ it both odd = nm

30. Obviously $\vec{A}\times\vec{C}$ and $\vec{B}\times\vec{D}$ will be non-zero collinear vectors

$\Rightarrow (\vec{A}\times\vec{C}).(\vec{B}\times\vec{D})\neq 0$ and similarly $(\vec{A}\times\vec{B}), (\vec{C}\times\vec{D})$ will be non-zero collinear vectors

Hence $(\vec{A}\times\vec{B})\times(\vec{C}\times\vec{D})=0$